

Quality-Critical Azure Lakehouse

DEMONSTRATED

From batch and streaming telemetry to governed, traceable data products.



Trust has to survive the whole data path

DEMONSTRATED

Operations needs to know whether a batch stayed within limits, what changed, which records failed, and where every KPI originated.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

CONTROLLED SEED 20260812

02 / 32

Two ingestion modes. One governed lakehouse.

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ADF lands deterministic batch files while Event Hubs feeds bounded telemetry into checkpointed Structured Streaming.



The evidence spans the engineering lifecycle

DEMONSTRATED

Provision, secure, ingest, transform, govern, orchestrate, break, repair, measure, monitor, publish, and tear down.

IMPLEMENTATION EVIDENCE

Bicep infrastructure as code**ADLS Gen2 medallion storage****Azure Data Factory batch ingestion****Event Hubs bounded streaming****PySpark and Delta Lake transformations****Lakeflow pipelines and Jobs****Unity Catalog governance and lineage****CDC and SCD Type 2 history****Failure repair and idempotency proof**

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04 / 32

Trial-only infrastructure with fail-closed cost gates

VERIFIED

Subscription-scope Bicep deployed the isolated resource group and reconciled Azure service set after a sanitized what-if.

BICEP / SUBSCRIPTION SCOPE

One bounded resource group, composed by modules

Trial-only infrastructure, scoped identity, diagnostics, budget, and explicit outputs.

TOP-LEVEL RESOURCES

9

reconciled inventory

REGION

EAST US 2

single deployment region

WORKSPACE SKU

TRIAL

no paid fallback

WHAT THE RECEIPT PROVES

main.bicep	subscription scope + budget-first dependency
storage.bicep	ADLS Gen2 / Standard_LRS / six containers
data-factory.bicep	managed identity + batch factory
event-hubs.bicep	Standard / 1 throughput unit / bounded stream
databricks.bicep	Trial workspace + Access Connector
monitoring.bicep	Log Analytics + diagnostics + alert
rbac.bicep	resource-group-scoped assignments

infra/main.bicep + evidence/public/receipts/resource-inventory.json

Deterministic build from public source artifacts

Managed identity is the default trust path

VERIFIED

ADF, the Access Connector, and federated deployment automation used scoped access without long-lived cloud passwords.

IDENTITY / PASSWORDLESS BY DEFAULT

EXECUTED / SANITIZED

Four trust paths, zero long-lived cloud passwords

ADF, Access Connector, GitHub Actions, and operators use scoped federated or managed identity.

LONG-LIVED TOKENS

0

public configuration readback

OIDC DEPLOYMENT

SUCCESS

resource-group scope

STORAGE KEYS USED

NO

source upload via Azure AD

WHAT THE RECEIPT PROVES

Azure Data Factory	system-assigned identity → ADLS landing
Access Connector	managed identity → Unity Catalog external locations
GitHub Actions	federated OIDC → isolated resource group
Databricks automation	Azure CLI authentication
Event Hubs credential	Key Vault-backed secret scope
Public evidence	tenant, subscription, principal IDs excluded

evidence/public/receipts/platform-configuration.json • source-upload.json

Deterministic build from public source artifacts

ADLS Gen2 separates landing, checkpoints, quarantine, and evidence

VERIFIED

Hierarchical namespace, TLS 1.2, disabled shared keys, and executed paths make state and ownership clear.

ADLS GEN2 / DELIBERATE STATE

EXECUTED / SANITIZED

Data, checkpoints, quarantine, and evidence are separated

Hierarchical namespace and managed identity make ownership and replay boundaries inspectable.

CONTAINERS

6

source, landing, managed, quarantine, checkpoints, evidence

FILES LANDED

6

commit-pinned batch inputs

AUTHENTICATION

AZURE AD

no storage keys

WHAT THE RECEIPT PROVES

source/	governed deterministic fixtures
landing/quality/	ADF file-level batch arrivals
managed/	Unity Catalog external data root
quarantine/	reason-coded rejected records
checkpoints/	Structured Streaming replay state
evidence/	governed run receipts
Security	HNS / TLS 1.2 / Standard_LRS / shared keys disabled

evidence/public/receipts/source-upload.json • infra/modules/storage.bicep

Deterministic build from public source artifacts

MICHAEL SETH WILLIAMS

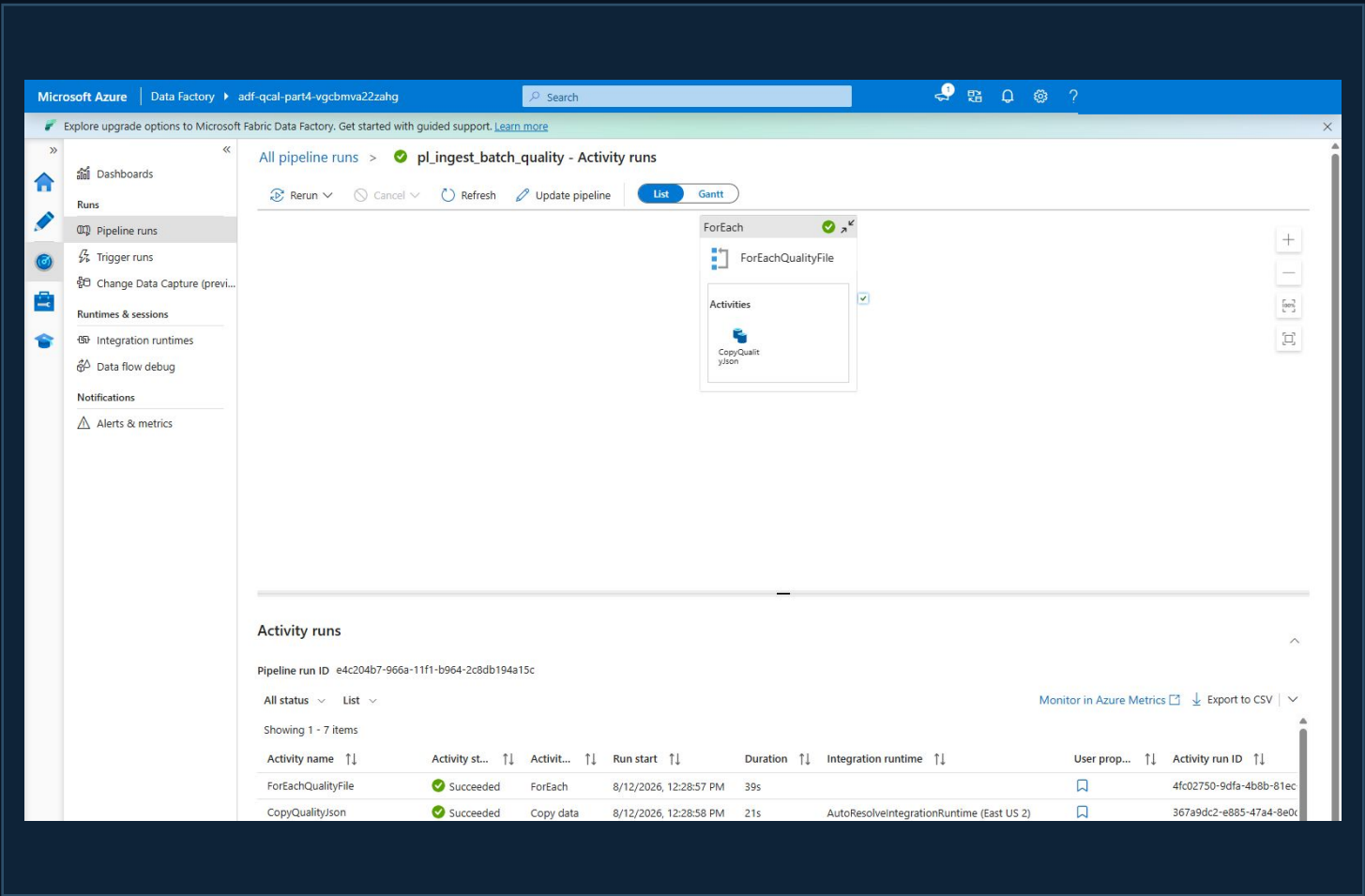
QUALITY-CRITICAL AZURE LAKEHOUSE

07

ADF ingested six commit-pinned source files

VERIFIED

The executed ForEach and Copy pattern preserves file-level transfer and activity evidence.



Six files and 30,000 rows reconcile

VERIFIED

The ADF run binds six successful Copy activities to 30,000 landed rows and byte-level receipts.

Microsoft Azure

Data Factory

adf-qcal-part4-vgcbmva22zahg

Search

Explore upgrade options to Microsoft Fabric Data Factory. Get started with guided support. [Learn more](#)

Dashboards

Runs

Pipeline runs

Trigger runs

Change Data Capture (previ...

Runtimes & sessions

Integration runtimes

Data flow debug

Notifications

Alerts & metrics

Rerun

Cancel

Refresh

Update pipeline

List

Gantt

Activity runs

Pipeline run ID e4c204b7-966a-11f1-b964-2c8db194a15c

All status

List

Monitor in Azure Metrics

Export to CSV

Showing 1 - 7 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID
ForEachQualityFile	Succeeded	ForEach	8/12/2026, 12:28:57 PM	39s			4fc02750-9dfa-4b8b-81ec-3f
CopyQuality/json	Succeeded	Copy data	8/12/2026, 12:28:58 PM	21s	AutoResolveIntegrationRuntime (East US 2)		367a9dc2-e885-47a4-8e0d-fi
CopyQuality/json	Succeeded	Copy data	8/12/2026, 12:28:58 PM	19s	AutoResolveIntegrationRuntime (East US 2)		e8f3faba-21ea-4359-87a8-b5
CopyQuality/json	Succeeded	Copy data	8/12/2026, 12:28:58 PM	16s	AutoResolveIntegrationRuntime (East US 2)		58b69df8-bc0b-4938-9607-b
CopyQuality/json	Succeeded	Copy data	8/12/2026, 12:28:58 PM	18s	AutoResolveIntegrationRuntime (East US 2)		1db1a118-b748-48aa-b92a-f
CopyQuality/json	Succeeded	Copy data	8/12/2026, 12:29:15 PM	18s	AutoResolveIntegrationRuntime (East US 2)		c7758225-9c2f-47dd-ad3a-9i
CopyQuality/json	Succeeded	Copy data	8/12/2026, 12:29:17 PM	15s	AutoResolveIntegrationRuntime (East US 2)		5079f566-b7d0-4006-9853-7

MICHAEL SETH WILLIAMS

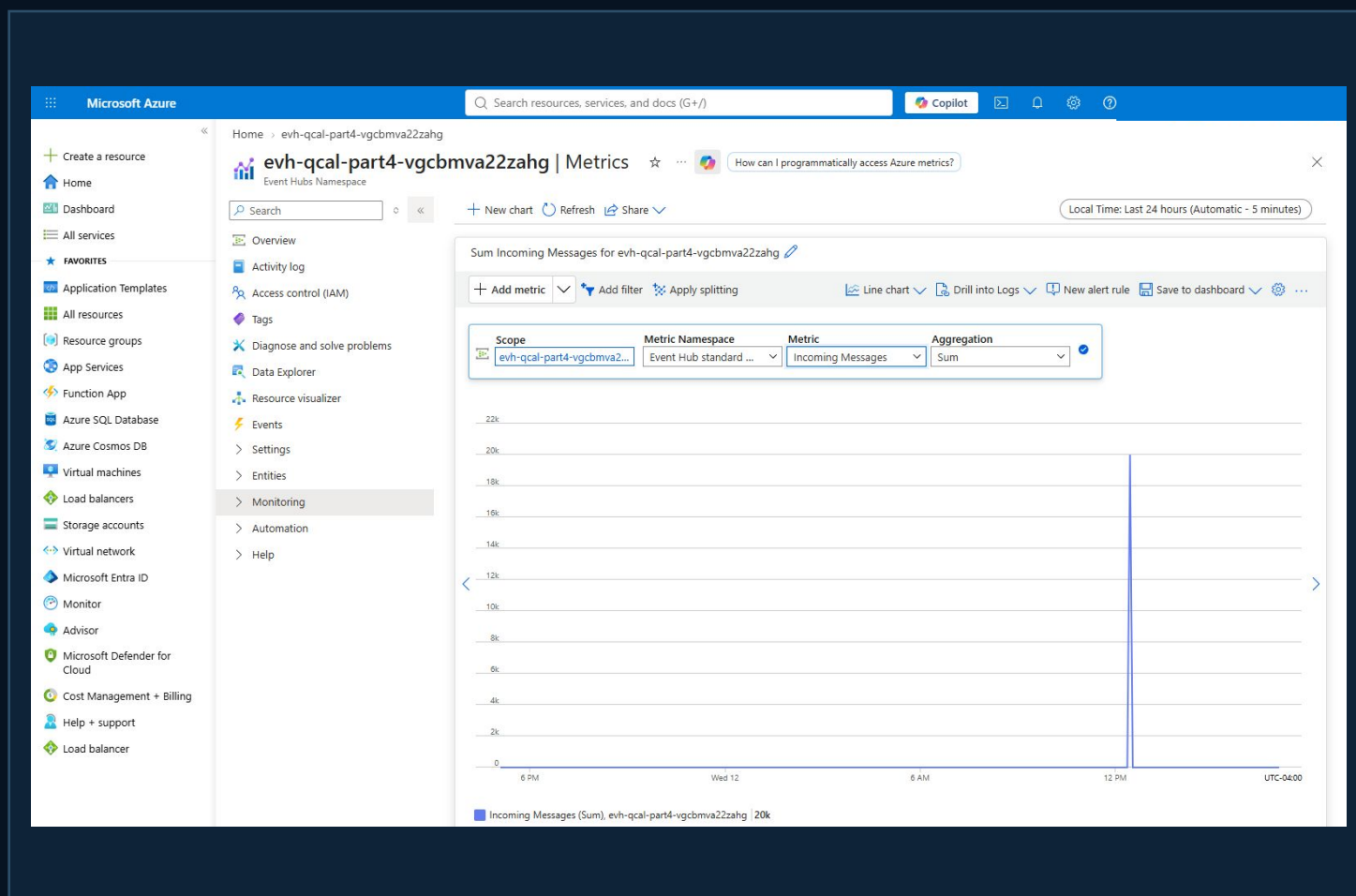
QUALITY-CRITICAL AZURE LAKEHOUSE

09

Twenty thousand deterministic telemetry messages

VERIFIED

Azure Monitor, the bounded producer receipt, final offsets, and Bronze rows all reconcile to 20,000.



Explicit schema, checkpoint, metadata, append semantics

VERIFIED

Event Hubs ingress, final offsets, zero lag, checkpoint state, and Bronze Delta counts reconcile.

STRUCTURED STREAMING / KAFKA OFFSETS

EXECUTED / SANITIZED

VERIFIED

PRODUCER, OFFSETS, CHECKPOINT, AND BRONZE ALL RECONCILE

Explicit schema and append semantics consumed the bounded Event Hubs stream into Delta.

PARTITION 0

10,198

final offset

PARTITION 1

9,802

final offset

BRONZE ROWS

20,000

Delta output

WHAT THE RECEIPT PROVES

Source	Kafka-compatible Event Hubs interface
Schema	explicit telemetry StructType
Write mode	append to Unity Catalog Delta table
Checkpoint	governed-volume/structured-streaming/bronze-telemetry
Maximum offset lag	0
Source metadata	partition, offset, event time, ingest time
Reconciliation	10,198 + 9,802 = 20,000 Bronze rows

evidence/public/receipts/structured-streaming-progress.json

2026-08-12T19:38:15.400895Z

Governance belongs inside the data path

VERIFIED

Catalog-qualified objects, managed storage, ownership, grants, tags, comments, masking, and external locations were applied.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

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12 / 32

Raw does not mean unaccountable

VERIFIED

Executed Bronze tables preserve source, timestamps, pipeline run, record hash, and schema version.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

CONTROLLED SEED 20260812

13 / 32

Allow, quarantine, and fail are different decisions

VERIFIED

Lakeflow published expectation metrics, routed rejected rows, and enforced one real fail-on-violation contract.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

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14 / 32

Bad records stay explainable

VERIFIED

Six quality rows and 17,144 telemetry rows remained reason-coded and inspectable rather than disappearing.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

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15 / 32

Conformance creates a dependable analytical boundary

VERIFIED

Explicit types, keys, units, domains, joins, deduplication, and idempotency shaped the executed Silver tables.

PYSPARK / SILVER CONTRACT

REVIEWABLE CODE

Validation routes every record deliberately

DEMONSTRATED

Reusable transformation code separates accepted rows from explainable quarantine.

QUALITY INPUT

30,000

Bronze rows executed

VALID

29,994

conformed Silver rows

QUARANTINED

6

reason-coded records

IMPLEMENTATION EXCERPT / `src/qcal/transforms.py`

```
44 def validate_quality(bronze: DataFrame, batch_master: DataFrame) -> ValidationResult:
45     """Apply explicit quality rules and route every invalid record to quarantine."""
46
47     known_batches = batch_master.select(F.col("batch_id").alias("_known_batch_id")).distinct()
48     duplicate_window = Window.partitionBy("record_hash").orderBy(
49         "source_file", "pipeline_run_id", "ingest_timestamp_utc"
50     )
51     expected_unit = (
52         F.when(F.col("test_type").isin("assay", "purity", "moisture"), F.lit("%"))
53         .when(F.col("test_type") == "bioburden", F.lit("CFU/g"))
54         .otherwise(F.lit(None).cast("string"))
```

Code: `src/qcal/transforms.py` • Validation: `tests/integration/test_medallion_transforms.py`

2026-08-12T20:37:18.32311Z

AUTO CDC preserves history instead of overwriting it

VERIFIED

The Lakeflow CHANGE flow upserted 647 effective-dated rows and passed the one-current-version invariant.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

CONTROLLED SEED 20260812

17 / 32

Six outputs with explicit business grain

VERIFIED

Two facts, three dimensions, and a site-product-date KPI table were published and validated.

IMPLEMENTATION EVIDENCE

fact_batch_quality

fact_cold_chain_excursion

dim_batch_history

dim_site

dim_product

kpi_quality_summary

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18 / 32

The DAG exposes dependencies and recovery boundaries

VERIFIED

Parallel Bronze paths converged at quality before Silver, AUTO CDC, Gold, validation, and receipt publication succeeded.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

CONTROLLED SEED 20260812

19 / 32

Gold is traceable to conformed source data

VERIFIED

Unity Catalog recorded the live relationship from Silver batch quality into the Gold fact and downstream consumers.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

CONTROLLED SEED 20260812

20 / 32

Traceability reaches the field level

VERIFIED

The selected path maps Silver result_value into Gold result_value and its derived quality_pass flag.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

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21 / 32

A reserved hard violation broke the real task

VERIFIED

The deterministic incident failed the live quality gate on reserved_schema_contract and HARD-FAIL-0001.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

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22 / 32

The platform named the violated contract

VERIFIED

The diagnostic connected schema version 99.0-reserved-failure to the affected quality flow and input record.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

CONTROLLED SEED 20260812

23 / 32

Repair the failed path, preserve completed work

VERIFIED

Lakeflow repair reran Bronze batch and dependent tasks while successful landing and streaming tasks remained one-attempt work.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The deterministic artifact is available for inspection.

CONTROLLED SEED 20260812

24 / 32

Recovery is verified by content, not appearance

VERIFIED

Counts, duplicates, aggregates, current-version checks, and clean-versus-recovered hashes all reconciled.

RECOVERY / CONTENT PROOF

EXECUTED / SANITIZED

Clean and repaired outputs are byte-for-byte equivalent

Counts, aggregates, duplicate checks, current-version invariant, and canonical hashes all reconcile.

VALIDATION

PASS

clean versus repaired

DUPLICATE KEYS

0

facts and KPIs

UPSTREAM RERUN

NO

successful tasks preserved

WHAT THE RECEIPT PROVES

Clean canonical SHA-256	85cb97df0b812676d91fba3935c58fcfb0de54f4dc92b23ddcdd71e0a222e806
Recovered canonical SHA-256	85cb97df0b812676d91fba3935c58fcfb0de54f4dc92b23ddcdd71e0a222e806
Expected / recovered match	TRUE
Current-version violations	0
Focused rerun task	bronze_batch
Dependent tasks rerun	TRUE
Outcome	deterministic repair without duplicate processing

evidence/public/receipts/lakeflow-recovery-validation.json

2026-08-12T21:00:20.3183821Z

A separate five-million-row fixture creates an honest test

VERIFIED

Three baseline runs recorded wall time and plan hashes; Spark Connect did not expose classic task and shuffle status-store metrics.

SPARK METRICS / BASELINE

EXECUTED / SANITIZED

Shuffle join with explicit repartition

Three complete executions on the same runtime and single-node compute.

MEDIAN WALL TIME

2.324 s

median of three

MEDIAN TASKS

NOT EXPOSED

Spark status store

MEDIAN SHUFFLE WRITE

NOT EXPOSED

runtime did not expose

WHAT THE RECEIPT PROVES

Run 1	5.125 s • — stages • — tasks • SHUFFLE
Run 2	2.324 s • — stages • — tasks • SHUFFLE
Run 3	2.239 s • — stages • — tasks • SHUFFLE
Result SHA-256	3eb397abd66bb16aec89b4703ab0157a612ac94f...
Physical plan	58ad05509e82ede192797f8c7a359f93d48b88a9...
Metric caveat	shuffle/spill not exposed by runtime status store

evidence/public/receipts/spark-performance-comparison.json

2026-08-12T21:08:45.678221Z

The justified optimization was slower

VERIFIED

Three broadcast runs matched result hashes but were 39.009% slower by median wall time on the same compute.

SPARK / THREE BY THREE

EXECUTED / SANITIZED

Measured optimization on five million skewed rows

VERIFIED

Both implementations produced the same result hash on identical compute.

BASELINE MEDIAN

2.324 s

3 runs / same compute

OPTIMIZED MEDIAN

3.230 s

broadcast + no repartition

WALL-TIME CHANGE

+39.009%

published without cherry-picking

WHAT THE RECEIPT PROVES

Fixture	5,000,000 rows / 80% hot key
Dimension	25 rows
Baseline	shuffle hash join + unnecessary repartition
Optimized	broadcast hash join + original partitioning
Result hashes match	TRUE
Baseline run times	5.125 s / 2.324 s / 2.239 s
Optimized run times	3.526 s / 3.127 s / 3.230 s

evidence/public/receipts/spark-performance-comparison.json

2026-08-12T21:08:45.678221Z

Configuration and observed outcomes stay distinct

VERIFIED

Diagnostics preceded workloads; no Log Analytics rows were visible in-window, and the data-plane failure did not fire the control-plane alert.

LOG ANALYTICS / LIVE TABLE DISCOVERY

EXECUTED / SANITIZED

VERIFIED

Queries used the tables that actually ingested

Diagnostics were configured before workloads; public output excludes workspace identifiers.

DISCOVERED TABLES

0

search * / Type summary

DIAGNOSTICS ORDER

BEFORE

relative to workloads

ALERT RULE

ENABLED

configuration readback

WHAT THE RECEIPT PROVES

Live query result

No records visible inside the bounded billing/ingestion window

Query

search * | summarize record_count=count() by Type

Identifier policy

workspace and customer IDs excluded

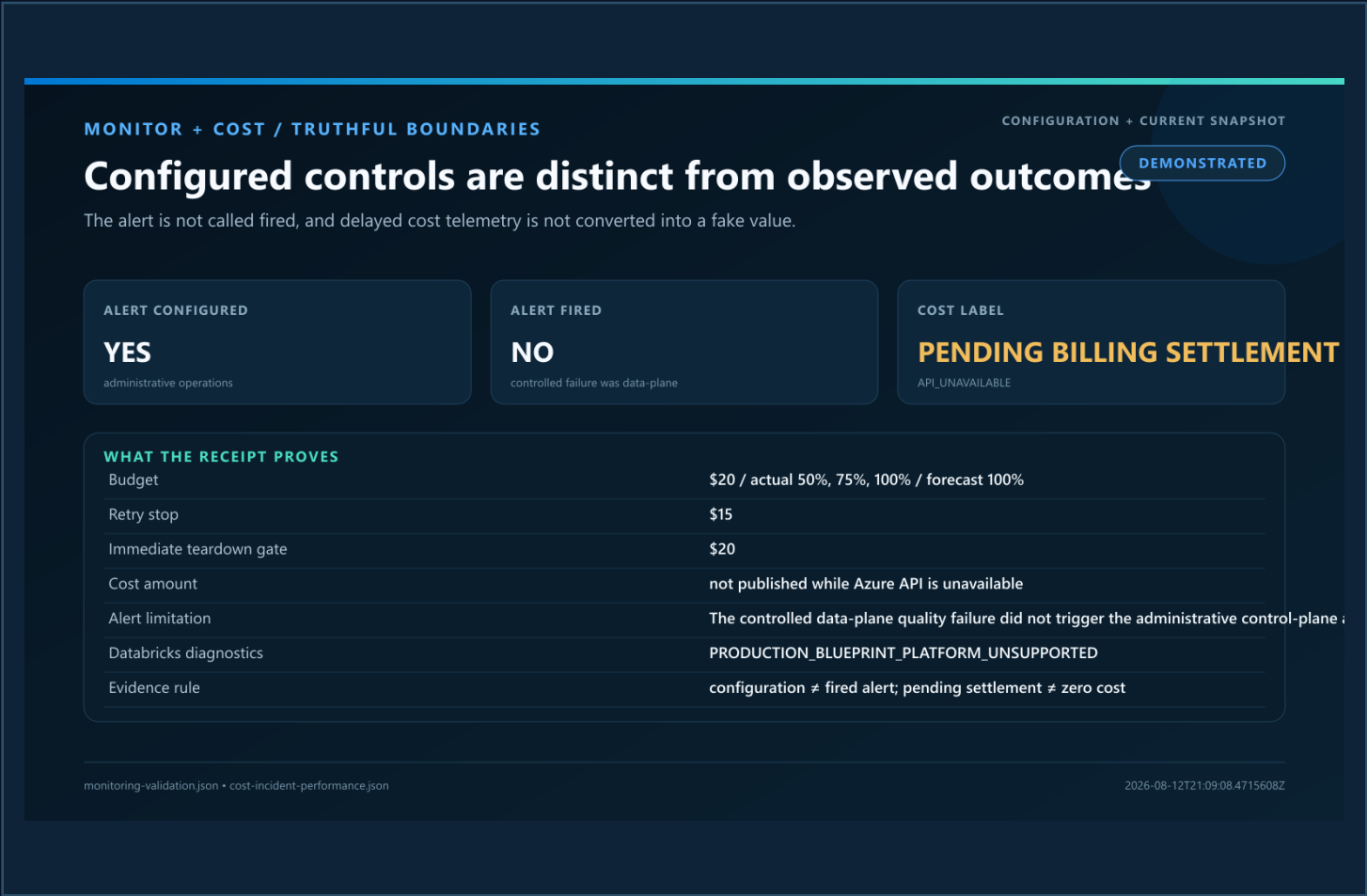
evidence/public/receipts/monitoring-validation.json

2026-08-12T21:25:52.1714090Z

Cost labels state exactly what billing can prove

DEMONSTRATED

The \$20 budget and stop gates were active; unavailable cost API telemetry remains PENDING BILLING SETTLEMENT, not zero.



Every release travels through validation

VERIFIED

Python, Spark, Bicep, bundle, content, accessibility, secret, metadata, PDF, Pages, and OIDC checks protect publication.

GITHUB ACTIONS / FEDERATED RELEASE

EXECUTED / SANITIZED

CI and Azure OIDC both passed without passwords

The release receipt binds workflow conclusions to public commit references.

CI
SUCCESS
run 31638712486

OIDC DEPLOY
SUCCESS
run 31618091562

LONG-LIVED SECRET
NONE
federated managed identity

WHAT THE RECEIPT PROVES

CI scope	Python / Spark / Bicep / bundle / site / PDF / publication scans
OIDC scope	read and write validation only inside rg-qcal-part4-dev
Repository	smailliwhtes/quality-critical-azure-lakehouse
CI commit	e67ceee1a743d047f4f44bfe78a87cbfc40b7856
CI conclusion	SUCCESS
OIDC conclusion	SUCCESS
Credential boundary	no client secret or Databricks token created

evidence/public/receipts/release-validation.json

2026-08-12T20:41:01Z

The lifecycle ends with authoritative absence

PRODUCTION_BLUEPRINT

Only the isolated resource groups and Part 4 budget are deleted, then Azure is polled until absence is confirmed.

EXECUTION EVIDENCE PENDING



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

Tests and reconciliation define what must be true.



EXECUTION RECEIPT

The platform artifact will promote this claim only after execution.

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31 / 32

Inspect the complete engineering journey

DEMONSTRATED

Every major claim connects to platform capture, implementation, receipt, validation, time, commit, and hash across the repository, live case study, evidence explorer, and teardown record.

IMPLEMENTATION EVIDENCE



IMPLEMENTATION

Code and configuration point to the exact engineering path.



VALIDATION

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EXECUTION RECEIPT

The deterministic artifact is available for inspection.

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32 / 32

<https://smailliwhtes.github.io/quality-critical-azure-lakehouse/>